

7-5 Quality Metrics

The following metrics are used to evaluate UPPMS quality after testing.

Metric	Definition	Measurement in the Project
Test Coverage (functional)	Share of critical modules/features covered by tests	Core modules covered: XML, Proposals, Tracks, Committees, GA, AI
Defect Density	Defects found ÷ component size	Defects logged per Spec Kit phase and re-tested
Pass Rate	Passed cases ÷ total executed cases	Delivery target: $\geq 95\%$ on critical paths
Response Time	API latency for common operations	CRUD/proposals feel interactive in UI; GA duration grows with input size
Error Rate	Share of incorrect responses / unhandled exceptions	Monitored via failing tests and Laravel logs
Security Isolation	No cross-university data leakage	Verified via TenantScope and multi-university Feature tests

Quality Acceptance Criteria

- No failures in security, isolation, or XML registration tests.
- Successful basic acceptance scenarios for Admin / Supervisor / Student.
- Ability to generate and approve a defense schedule on Campus Demo without visible hard conflicts.